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Manufacturing Processes - II (ME 245)

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ME 245 Manufacturing Processes II	
CO 1	Understand working of lathe, shaper and planer, drilling, milling and grinding machines.
CO 2	Comprehend speed and feed mechanisms of machine tools.
CO 3	Study and practice on machine tools and their operations
CO 4	Estimate machining times for machining operations on machine tools.
CO 5	Gain knowledge of classification of metal forming processes, plastic deformation, flow stress, formability, strain, effect of temperature, strain rate and metallurgical structure on metal working, friction and lubrication, deformation zone geometry, workability, and residual stresses.
CO 6	Understand metal forming processes like forging, drawing, rolling, extrusion, and deep drawing.

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EXPERIMENT: 1

Lathe Machine: Turning and Thread Cutting

AIM: To study about turning operation and thread cutting operation

APPARATUS: Lathe Machine, Taper Turning Attachment, Tooling.

OBJECTIVE:

- To know the construction and features of lathe machine
- To perform plain and taper turning on the lathe
- To perform thread cutting operation on lathe machine
- To know the difference between right and left handed threads

THEORY:

Lathe is the oldest machine tool. It is a versatile machine tool, which can perform variety of operations like plain turning, taper turning, threading, facing, boring, drilling, reaming, parting off etc. The workpiece is held securely by appropriate device (usually chuck) on lathe and cutting tool is fed against the rotating job to remove material in the form of chips.

For doing different operations over lathe machine, different mechanisms are assembled on it, which render different function to accomplish the job. Some examples are as listed below

- Head stock mechanism for changing the speed of rotating job
- Carriage mechanism for giving longitudinal & traverse motion to the tool held in tool post.
- Feed gearbox & Apron mechanism for transmitting motion from headstock spindle to carriage & simultaneously for providing required feed to the tool.
- Thread cutting mechanism for cutting the threads of required pitch.

Lathe Accessories:

They are holding and supporting devices for either workpiece or tool like lathe centers, catch & carrier plates, chucks, collets, face plates, mandrel, rests etc.

Lathe Attachments:

These are additional equipments used for specific purposes like thread chasing dials, taper turning attachment, turret, grinding & milling attachments.

Turning:

Turning is the operation of removing excess material from the rotating workpiece to produce a cylindrical or cone-shaped surface.

Operation for producing cylindrical surface is called plain turning while for producing cone-shaped surface is called taper turning.

Taper is defined as uniform increase or decrease in diameter of a workpiece when measured along its length. Elements of taper turning are

D - Large diameter of job
d - Small diameter of job
l - Length of job along the axis.
α - half-taper angle.

Conicity, k = $\frac{(D-d)}{l}$

Tan α = $\frac{(D-d)}{2l}$

Methods of Taper Turning:

- Taper turning by form tool
- Taper turning by setting over tailstock.
- Taper turning by swiveling compound rest.
- Taper turning by taper turning attachment
- Taper turning by combined feed

Thread Cutting:

For Right Hand Thread cutting, carriage must move towards headstock and for Left Hand threads, Carriage must move towards tail stock, while Job rotates in the anti-clockwise direction looking form tail stock side.

Calculation:

Change gear calculations:-

These calculations are required for thread cutting & the formula is as under

$$\frac{\text{NUMBER OF DRIVER TEETH}}{\text{NUMBER OF DRIVEN TEETH}} = \frac{\text{Lead screw turn}}{\text{Spindle turn}} = \frac{\text{Pitch of the Thread to be cut}}{\text{Pitch of the lead screw}}$$

Set of gears is available from 20 teeth to 120 teeth in steps of 5 teeth. One gear is supplied with 127 teeth as translating gear. Pair of intermediate gear or idle gears area assembled in between driver & driven gears for same direction of rotation and is fixed as per standard practice.

Metric threads of British lead screw is specified by TPI (Threads per Inch) instead of pitch, the calculations for it is as under

$$\frac{\text{NUMBER OF DRIVER TEETH}}{\text{NUMBER OF DRIVEN TEETH}} = \frac{\text{Lead screw turn}}{\text{Spindle turn}} = \frac{\text{Pitch of the Thread to be cut}}{\text{Pitch of the lead screw}}$$

Since, $\left[Pitch(In\,Inch) = \frac{1}{Thread\ Per\ Inch} \right]$

$$\frac{\text{DRIVER TEETH}}{\text{DRIVEN TEETH}} = \frac{\text{Pitch of the Workpiece (p)}}{\text{Pitch of the lead screw}\left(\frac{25.4}{n}\right)} = \frac{p}{\left(\frac{1}{n} \times \frac{127}{5}\right)} = \frac{5pn}{127}$$

Normally multi start threads are either two starts or three starts. The independent threads are called starts.

For 3 start threads: one complete turn of thread advances three times as far as if it was a single start thread. The distance a multiple start screw advances along its axis in one turn is called lead.

$$\frac{\text{NUMBER OF DRIVER TEETH}}{\text{NUMBER OF DRIVEN TEETH}} = \frac{\text{Lead of the Thread to be cut}}{\text{Pitch of the lead screw}} = \frac{\text{Pitch} \times \text{No. of start}}{\text{Pitch of the lead screw}}$$

Lead of thread = pitch of thread x No. of starts

In multi start threads, circumference of the job is divided equally as per no. of starts of the thread. Then every division becomes the starting point of thread.

Three methods of equally dividing the circumference of job are:

- Marking the gears
- Moving top slide
- Using an indexing drivers plate

REVIEW QUESTIONS:

1. List the specification criteria for a lathe machine.
2. Is there any difference between feed rod and lead screw? Explain.
3. What is Taper and conicity?
4. Describe the taper turning method used by you for making taper surface with figure.
5. What is the material of tool used for Taper Turning?
6. What is the relation between pitch, no. of start, & lead of the screw?
7. What do you mean by three start threads?
8. At what degrees job circumference is divided for making 2 & 3 starts threads?
9. What are the different methods used for dividing the circumference of the job for making multi start threads? In addition, describe briefly the method used in practical for making three start threads.
10. What will happen if job rotates on clockwise direction during threading?

Marks obtained:	**Signature of faculty:**	**Date:**

Manufacturing Processes - II (ME 245) **Date:**

EXPERIMENT: 2

Capstan and Turret Lathe

AIM: To study about capstan and turret lathe

APPARATUS: Capstan Lathe Machine.

OBJECTIVE:

- To get familiar with the parts of capstan & turret lathes
- To know the advantages and function of turret

THEORY:

The capstan & turret lathes are fundamentally production machines, capable of producing large number of identical pieces in a minimum time. In a capstan or turret lathe, there is no tailstock, but in its place a hexagonal turret is mounted upon a slide, which rest on the bed. All the six faces of the turret can hold six or more number of different tools. The turret may be indexed automatically and each tool may be brought in line with the lathe axis in a regular sequence.

The size of capstan or turret lathe is designated by the maximum diameter of rod that can be passed through the headstock spindle and swing diameter of the work that can rotate over the lathe bed ways. Other particulars such as number of spindle speeds, number of feed available to the carriage, floor space, and power required could also be stated to specify the lathe fully.

Capstan or Ram Type Lathe:

It carries the hexagonal turret on a ram or a short slide. The ram slides longitudinally on a saddle positioned and clamped on lathe bed ways. This type of machine is lighter in construction and is suitable for machining bar of smaller diameter.

Turret or Saddle Type Lathe:

It carries the hexagonal turret directly on a saddle and whole unit moves back and forth on the bed ways to apply feed. This type of turret is heavier in construction and is particularly adapted for large diameter bar work and chucking work. The machine can accommodate longer workpiece than that in capstan lathe.

REVIEW QUESTIONS:

1. State difference between a Capstan & Turret lathe and an engine lathe.
2. Distinguish between Capstan lathe and Turret lathe.
3. List the specification criteria for a Capstan and Turret lathe.

Marks obtained:	Signature of faculty:	Date:

EXPERIMENT: 3

Drilling Machine

AIM: To study about Drilling Machine
APPARATUS: Radial Drilling Machine.
OBJECTIVE:

- To study the construction of drilling machine
- How make different size of holes using drilling machine
- To study the different size of drill bits and drill geometry

THEORY:
The main function of drilling machine is to generate hole, quickly & at low cost. The hole is generated by rotating edge of a cutting tool known as drill which exerts large force on the work clamped on table. As the machine exerts vertical pressure to generate hole, it is called as drill press.

Drilling Machine Classification:

1. Hand Drill Machine
2. Portable Drilling Machine
3. Sensitive Drilling Machine
4. Upright Drilling Machine
   - Round Column Section / Pillar Type Drilling Machine
   - Box Column Section Drilling Machine
5. Radial Drilling Machine
   - Plain Type
   - Semi-Universal Type
   - Universal Type
6. Gang Drilling Machine
7. Multiple Spindle Machine.
8. Automatic Drilling Machine
9. Deep Hole Drilling Machine

Drilling Machine Size & Specifications:

Machine Type	**Specified By**
Portable drill	Maximum size of drill it can hold
Sensitive & upright drilling machine	largest diameter of job in which machine can drill a center hole & maximum

	drilling capacity.
Radial drilling machine	Length of arm & sometimes size of column
Multi spindle drilling machine	Maximum drilling capacity, largest Drill it can hold & total drilling area.

Different Operations done on Drilling Machine:

1. Drilling
2. Reaming
3. Boring
4. Counter boring
5. Counter shrinking
6. Spot facing
7. Tapping
8. Lapping
9. Grinding
10. Trepanning

Selection of Cutting Speed (v) & Feed (f) Depends Upon:

1. Material of job.
2. Cutting tool material
3. Surface finish required
4. Efficient use of cutting fluid
5. Work holding method
6. Size, type & rigidity of machine

REVIEW QUESTIONS:

1. List various drilling machines that we have in our workshop and describe machine that you have used for drilling operation.
2. State different method of holding Drill in a Drilling machine Spindle and describe Tool holding device that you have used in the practical.
3. Which work holding device you have used in the practical?
4. What are the materials of drill?
5. Explain the back gear arrangement of drilling machine.
6. Show with neat sketches the constructional features of a twist drill and label the important features.
7. What are the different types of drills used? Explain the function of each of the type of drill.
8. Discuss the problems faced in drilling operations with their causes and possible remedies.

9. A hole with 40 mm diameter and 50 mm depth is to be drilled in mild steel component. The cutting speed can be taken as 65 m/min and the feed rate as 0.25 mm/rev. Calculate the machining time and the material removal rate.

Marks obtained:	Signature of faculty:	Date:

EXPERIMENT: 4

Spur Gear Cutting on Milling Machine

AIM: to study about Milling Machine

APPARATUS: Milling Machine, Indexing Mechanism

OBJECTIVE:

- To study the construction of milling machine
- To cut spur gear with the help of a milling machine
- To study the indexing mechanism in milling machine

THEORY:

A milling machine is a machine tool that removes metal as the work is feed against a rotating multipoint milling cutter. The machine can also hold one or more number of cutters at a time.

Milling Cutters:

There are many different type of standard milling cutter available like plain milling cutter, side milling cutter, end milling cutter, fly cutter, formed milling cutter, tap & reamer cutter, metal slitting saw, angle milling cutter, t-slot cutter. Using different types of cutter various operations can be carried out.

Spur Gear Cutting:

Gears whose axis and teeth are parallel to the center line of the gear are called spur gears. The specifications include Addendum, Dedendum, Clearance, Face of tooth, Flank of tooth, Tooth surface, Thickness, Working depth, outside circle, Pitch circle, Root circle, Pitch point, Pitch line, Circular pitch, Module (in), Pressure angle etc. Many times diametral pitch (p) is used instead of the module. Diametral pitch is reciprocal of module.

Indexing:

Indexing is dividing the circumference of job into required numbers of equal parts. An indexing head, also known as a dividing head or spiral head, is a specialized tool that allows a work piece to be circularly indexed; that is, easily and precisely rotated to pre-set angles or circular divisions. Indexing heads are usually used on the tables of milling machines.

Direct Indexing Plate:

Most dividing heads have an indexing plate permanently attached to the spindle. This plate is located at the end of the spindle, very close to where the work would be mounted. This plate is usually equipped with a series of holes that enables rapid indexing to common angles, such as 30, 45, or 90 degrees. A pin in the base of the dividing head can be extended into the direct indexing plate to lock the head quickly into one of these angles.

REVIEW QUESTIONS:

1. Enlist different types of milling operations and explain about any three in detail.
2. What is the basic feature of Universal Milling Machine?
3. Classify milling cutters and sketch any four standard milling cutters stating their applications.
4. What do you mean by Indexing? Explain indexing method used for gear cutting on milling machine.
5. Differentiate between up milling and down milling.
6. Differentiate between compound indexing and differential indexing.
7. Explain the construction of dividing head.
8. A C50 steel flat surface of dimensions 100 mm x 250 mm is to be produced on a horizontal axis milling machine. An HSS slab mill with a 100 mm diameter and 150 mm width is to be used for the purpose. The milling cutter has 8 teeth. Calculate the machining time assuming that entire stock can be removed in one depth of 2 mm.
9. Write down the name of any five component in which milling operation is required.

Marks obtained:	**Signature of faculty:**	**Date:**

EXPERIMENT: 5

Shaping and Planing Machine

AIM: To study about Shaping and Planing Machine.
APPARATUS: Shaping & Planing Machine, Tooling.
OBJECTIVE:

- To get acquainted with shaping & planing machine operations.
- To know the differences in mechanisms of shaper, planar and slotter.

THEORY:
Shaping machine employs reciprocating motion of cutting tool for machining flat surfaces. Single point cutting tool is held securely in Tool post, which is mounted on ram. Work piece is held in Vise, which is mounted, on shaper table. The ram reciprocates and so the cutting tool. The job held in vise is given traverse feed while tool is reciprocates and hence cutting action is achieved.

The travel of ram is called stroke. The shaper is specified mainly on the length of stroke apart from other details such as, type of drive, power input, cutting to return stroke ratio, feed, floor space requirements etc.

The planer is similar to a shaper in the sense that it is mainly used to produce plane & flat surfaces using a single point tool, but as the work is supported on a reciprocating table and feed is given to a stationary tool. Planer is very large & heavy in construction. It is mainly used for machining heavy work pieces which cannot be machined on shaper due to limitations on ram stroke length

Shaper Mechanism:
The rotary motion of the shaper drive is converted in to reciprocating motion by suitable mechanism. The reciprocating motion of ram is divided in to two parts, forward stroke and return stroke. The cutting action is accomplished during forward stroke while return stroke is less compared to that of forward stroke. The mechanism imparts such action is called Quick Return Mechanism. There are different methods by which such action can be achieved, such as...

1. Crank and slotted link mechanism
2. Whitworth quick return mechanism
3. Hydraulic shaper mechanism

Other important mechanism of shaper is fed mechanism for imparting automatic feed to job.

Types of Planning Machines:

1. Double housing planer.
2. Open side planer.
3. Pit planer.
4. Edge or Plate planer.
5. Divided Table planer.

Size of Planer:

A. **Standard planer** is specified by the size of the largest rectangular solid that can reciprocate under the tool (i.e. distance between the two housings, the height from top of the table to the upper most position of cross rail and the maximum length of table travel.)
B. **Open side planer** is specified by the size of the largest Job that can be machined on its table. Maximum width of the job (i.e. Planing width) is the distance from tableside of column to the tool in the outer tool head (which extends beyond table width by approx 300 mm) in a vertical position.

Other particular like no. of speeds & feeds available, power input, floor space required, net weight, type of drive are also specified.

Important Planer Mechanisms:

1. Table drive mechanism
    a. Open & cross belt drive
    b. Reversible motor drive
    c. Hydraulic drive
2. Feeding mechanism —usually by friction disc.

Common Planer Operations:

1. Planing flat horizontal surfaces
2. Planing vertical surfaces
3. Planing at an angle & machining dovetails
4. Planing curved surfaces
5. Planing slots & grooves

Slotting Machine:

The slotting machine is a reciprocating machine tool in which, the ram holding the tool reciprocates in a vertical axis and the cutting action of the tool is only during the downward stroke.

The slotting machine is used for cutting grooves, keys and slots of various shapes making regular and irregular surfaces both internal and external cutting internal and external gears and profiles. The different types of slotting machines are:

1. Punch slotter: a heavy duty rigid machine designed for removing large amount of metal from large forgings or castings

2. Tool room slotter: a heavy machine which is designed to operate at high speeds. This machine takes light cuts and gives accurate finishing.

3. Production slotter: a heavy duty slotter consisting of heavy cast base and heavy frame, and is generally made in two parts

REVIEW QUESTIONS:

1. Briefly explain the ram reciprocating mechanism for shaper machine used in practical.
2. Explain automatic feeding mechanism of shaper machine.
3. State the varieties of operations that can be performed on shaper machine.
4. State work holding devices used for shaper machine.
5. State the materials of shaper tool.
6. Distinguish between Vertical shaper and Slotting Machine.

Marks obtained:	Signature of faculty:	Date:

EXPERIMENT: 6

Grinding Machine

AIM: To study about surface grinding operation.
APPARATUS: Surface Grinding Machine, Tooling
OBJECTIVE:

- To know about grinding machine operation
- To know about various abrasive materials and grain size

THEORY:

Grinding is a metal cutting operation by using a rotating cutting tool called abrasive wheel. It is generally used to obtain high surface quality, accuracy of shape & dimension. The grinding machine consists of a power driven grinding wheel spinning at the required speed (which is determined by the wheel's diameter and manufacturer's rating, usually by a formula) and a bed with a fixture to guide and hold the work-piece. The grinding head can be controlled to travel across a fixed work piece or the workpiece can be moved whilst the grind head stays in a fixed position.

Grinding machines remove material from the workpiece by abrasion, which can generate substantial amounts of heat. In very high-precision grinding machines (most cylindrical and surface grinders) the final grinding stages are usually set up so that they remove about 200 nm (less than 1/100000 in) per pass - this generates so little heat that even with no coolant, the temperature rise is negligible.

Classification of Grinding Machines:

1. Rough or non precision grinder
    a. Floor stand and Bench grinder
    b. Portable and Flexible shaft grinder
    c. Swing frame grinder
    d. Abrasive belt grinder
2. Precision grinder
    a. External cylindrical grinder
        i. Cylindrical grinder, Surface grinder.
    b. Internal cylindrical grinding.
        i. Chucking, Planetary, Center less Grinders.
    c. Tool and cutter grinder
    d. Special grinder

REVIEW QUESTIONS:

1. List the grinding machines we have in our workshop.
2. On which grinding machine you have performed practical?
3. Give the designation of grinding wheel that you have used?
4. Explain the terms: Glazing, Loading, Dressing, Truing & Balancing.
5. What are the various abrasive machining operations? Explain all in brief.
6. What are the grinding process parameters? Explain their effect on the grinding performance.
7. What are the advantages and limitations of using centre less grinding?
8. Using a horizontal axis surface grinder, a flat surface of C65 steel of size 100 mm x 250 mm is to be ground. A grinding wheel with 250 mm diameter and 20 mm thickness is used. Calculate the grinding time required. Assume a table speed of 10 m/min and wheel speed of 20 m/s.

Marks obtained:	**Signature of faculty:**	**Date:**

EXPERIMENT: 7

Rolling Operation and Three-Roll Bending Machine

AIM: Study of rolling operation with three roll bending machine. Rolling of sheet and prepare cylinder using three-roll bending machine.

APPARATUS: Three-roll Bending Machine.

OBJECTIVES:

- To understand between bending & rolling of shell, plate & section.
- To know the metallurgical changes during rolling operations continuous & non-continuous.
- To know about different rolling processes & its classifications.
- Metal rolling mills, types & principles.
- Rolling variables & its design criteria.
- Defects in rolled products, inspection & remedies.

THEORY:

Rolling is a fabricating process in which the metal, plastic, paper, glass, etc. is passed through a pair (or pairs) of rolls. There are two types of rolling process, flat and profile rolling. In flat rolling the final shape of the product is either classed as sheet (typically thickness less than 3 mm, also called "strip") or plate (typically thickness more than 3 mm). In profile rolling the final product may be a round rod or other shaped bar, such as a structural section (beam, channel, joist etc). Rolling is also classified according to the temperature of the metal rolled. If the temperature of the metal is above its recrystallisation temperature, then the process is termed as hot rolling. If the temperature of the metal is below its recrystallisation temperature, the process is termed as cold rolling. Another process also termed as 'hot bending' is induction bending, whereby the section is heated in small sections and dragged into a required radius.

Heavy plates tend to be formed using a press process, which is termed forming, rather than rolling.

REVIEW QUESTIONS:

1. Write down different rolling processes & their classification with brief description of each.
2. Explain the principle of metal rolling mills.
3. Which are the rolling variables to keep under control for best product output?

4. Write down the defects in rolled products.
5. How does cold rolling differs from hot rolling?
6. What is meant by ‘grain flow’ and ‘breakdown passes’ in rolling?
7. Explain the meaning of draught and elongation as related to hot rolling.

Marks obtained:	**Signature of faculty:**	**Date:**

EXPERIMENT: 8

Forging Operation

AIM: Study of forging operation and generating square headed bolt using different forging operation.

APPARATUS: Open hearth furnace, Anvil, Hammer

OBJECTIVES:

- Definition of Forging.
- Material behaviour of different metals during forging.
- Classification of forging process & operation.
- To know the forging defects.

THEORY:

Forging is the term for shaping metal by using localized compressive forces. Cold forging is done at room temperature or near room temperature. Hot forging is done at a high temperature, which makes metal easier to shape and less likely to fracture. Warm forging is done at intermediate temperature between room temperature and hot forging temperatures. Forged parts can range in weight from less than a kilogram to 170 metric tons. Forged parts usually require further processing to achieve a finished part.

Hot forging is defined as working a metal above its recrystallisation temperature. The main advantage of hot forging is that as the metal is deformed the strain-hardening effects are negated by the recrystallisation process. Cold forging is defined as working a metal below its recrystallisation temperature, but usually around room temperature. If the temperature is above 0.3 times the melting temperature (on an absolute scale) then it qualifies as warm forging.

There are many different kinds of forging processes available; however they can be grouped into three main classes

- Drawn out: length increases, cross-section decreases
- Upset: Length decreases, cross-section increases
- Squeezed in closed compression dies: produces multidirectional flow

REVIEW QUESTIONS:

1. What is forging in general?
2. Write down all the forging processes & machines with brief description.
3. Specify forging defects & its remedies.

4. What do you understand by the term flash in forging? Explain with neat sketch.
5. Why are die inserts used in forging dies? Explain with example.
6. Write down the name of any five products which is manufactured by forging operation.

Marks obtained:	**Signature of faculty:**	**Date:**

EXPERIMENT: 9

Bending Operation and Hydraulic Pipe Bender

AIM: Study of bending operation. Pipe bending practice using hydraulic pipe bending machine, vertical and horizontal conduit pipe bender

APPARATUS: Manual Pipe Bending Machine, Hydraulic Pipe Bending Machine

OBJECTIVES:

- To know principle of bending.
- To know bending process
- To know effect of bending along & across the grain.

REVIEW QUESTIONS:

1. Explain the principle of bending.
2. How pipe bending can be done?
3. Which one is desirable, bending across the grain or along the grain? Why?
4. Compare manual pipe bending and hydraulic pipe bending operation with suitable application.
5. Write down any ten applications of pipe bending.

Marks obtained:	Signature of faculty:	Date:

EXPERIMENT: 10

Press Working Operations

AIM: To study various press working operations.
APPARATUS: Hydraulic ram, Die and Punch
OBJECTIVES:

- To know set up of press working. (tools)
- To know fundamental of shearing action.
- To know various press working operations.
- To know the principle and setup of drawing operation.
- To know various drawing operation

THEORY:

The basic operations divided into several categories such as *cutting, bending, forming, drawing* and *squeezing.* All of these operations require that portions of the work piece undergo *plastic deformation*. To accomplish this, pressures in excess of the *yield point* of the material are used. Understanding how these processes are appropriately applied, is a key designing and maintaining dies for successful press working. Knowledge of material and die interactions is also required for cost effective design, repair and troubleshooting.

Wire drawing is a metalworking process used to reduce the diameter of a wire by pulling the wire through a single, or series of, drawing die(s). There are many applications for wire drawing, including electrical wiring, cables, tension-loaded structural components, springs, paper clips, spokes for wheels, and stringed musical instruments. Although similar in process, drawing is different from extrusion, because in drawing the wire is pulled, rather than pushed, through the die. Drawing is usually performed at room temperature, thus classified a cold working process, but it may be performed at elevated temperatures for large wires to reduce forces.

REVIEW QUESTIONS:

1. List out different types of die used in sheet metal working.
2. Enlist various press working processes.
3. Explain the effect of shear on cutting.
4. Explain the principle of sheet metal shearing.
5. Differentiate between punching, piercing & blanking.
6. State & explain different drawing process & with sketch.
7. Explain rod, tube & wire drawing with neat sketch.

8. Write down the name of any five products which is manufactured by press working operation.

Marks obtained:	**Signature of faculty:**	**Date:**